Filling the grid

A container of fixed width and height $w, h \in \mathbb{R}^+$ is filled with $n \in \mathbb{N}$ (natural numbers without 0) squares positioned in a grid-like fashion. The squares should cover as much of the $w \times h$ area as possible without overflowing, which means we must maximize the side length $S \in \mathbb{R}^+$ of the square. In other words, we are searching a $b \times a \in \mathbb{N}^2$ grid which fits all our n squares and itself fits in the container while either being as wide or as tall as the container.

We can state the problem in a formal manner as follows:

$$\begin{gathered}
\mathrm{Sol}_a := \left\{ s \in \mathbb{R}^+ \;\middle|\; \exists a, b \in \mathbb{N} \,.\, n \leq ab \wedge as = w \wedge bs \leq h \right\} \\
\mathrm{Sol}_b := \left\{ s \in \mathbb{R}^+ \;\middle|\; \exists a, b \in \mathbb{N} \,.\, n \leq ab \wedge bs = h \wedge as \leq w \right\} \\
\mathrm{Sol} := \mathrm{Sol}_a \cup \mathrm{Sol}_b \\
S := \max \mathrm{Sol}
\end{gathered}$$

Let's start solving for a solution $s_a \in \mathrm{Sol}$ by attempting to find a grid as wide as the container. For any $s \in \mathrm{Sol}_a$:

$$\begin{aligned}
n \leq ab \wedge as = w \wedge bs \leq h &\implies n \leq \frac{w}{s} b \wedge bs \leq h \\
&\implies n \leq \frac{w}{s} \cdot \frac{h}{s} \\
&\implies n \leq \frac{wh}{s^2} \\
&\implies s \leq \sqrt{\frac{wh}{n}} \qquad (1)
\end{aligned}$$

This is an upper bound on the side length. Let's determine the grid's number of columns, a:

$$\begin{aligned}
as = w &\implies a = \frac{w}{s} \\
&\overset{(1)}{\implies} a \geq \frac{w}{\sqrt{\frac{wh}{n}}} \\
&\implies a \geq \sqrt{\frac{w}{h} n} \\
(s \text{ maximized} \wedge a \in \mathbb{N}) &\implies a = \left\lceil \sqrt{\frac{w}{h} n} \right\rceil \qquad (2)
\end{aligned}$$

To further clarify the last step: since s is in the denominator, maximizing it means a must take the smallest value greater than or equal to RHS. Since $a \in \mathbb{N}$ this is precisely RHS rounded up.

Knowing a we can now determine the side length s:

$$as = w \implies s = \frac{w}{a} \tag{3}$$

Since the implications in (1) go in one direction only, we must check that $s \in \mathrm{Sol}$. The first necessary condition is:

$$n \leq \left\lfloor \frac{w}{s} \right\rfloor \cdot \left\lfloor \frac{h}{s} \right\rfloor \tag{4}$$

since $a = \lfloor w/s \rfloor$ and $b = \lfloor h/s \rfloor$ provide the number of squares the grid *should* have on each dimension, if it fits the squares and doesn't overflow. Moreover, $as = w \implies a = w/s = \lfloor w/s \rfloor$, since $a \in \mathbb{N}$.

Let's go to our initial assumptions.

$$n \leq ab \land bs \leq h \implies n \leq a\frac{h}{s} \implies \underbrace{n \leq a\left\lfloor \frac{h}{s} \right\rfloor}_{\text{(i)}} \lor \underbrace{(n > a\left\lfloor \frac{h}{s} \right\rfloor \land n \leq a\frac{h}{s})}_{\text{(ii)}} \tag{5}$$

In case (i) we can take $b = \lfloor h/s \rfloor$, which means $bs \leq h$. Thus, s as determined above is in $\mathrm{Sol}_a \subseteq \mathrm{Sol}$. But in case (ii), $b = \lfloor h/s \rfloor \implies n > ab \implies s \notin \mathrm{Sol}$. A new solution s' must be found.

Let's analyze what follows from (ii):

$$\begin{aligned} a\left\lfloor \frac{h}{s} \right\rfloor < n \leq a\frac{h}{s} &\implies \left\lfloor \frac{h}{s} \right\rfloor < \frac{h}{s} \land n \leq a\frac{h}{s} \\ &\implies \frac{h}{s} < \left\lceil \frac{h}{s} \right\rceil \land n \leq a\frac{h}{s} \\ &\implies n < a\left\lceil \frac{h}{s} \right\rceil \end{aligned} \tag{6}$$

This means $b = \lceil h/s \rceil$ is a safe choice for the a that we've determined, and a candidate for s' is:

$$s' = \frac{h}{b} = \frac{h}{\lceil \frac{h}{s} \rceil} \overset{(3)}{=} \frac{h}{\lceil \frac{h}{w}a \rceil} \tag{7}$$

Is this $s' \in \mathrm{Sol}$? Yes, in fact in Sol_b: $(6) \implies n \leq ab$, $(7) \implies bs' = h$ and

$$as' = \frac{ah}{\lceil \frac{h}{w}a \rceil} = \frac{ah}{\lceil \frac{ah}{w} \rceil} \overset{\lceil x \rceil \geq x}{\implies} as' \leq \frac{ah}{\frac{ah}{w}} \implies as' \leq w \tag{8}$$

With that we have determined s_a. Here's its formula based only on inputs w, h, n:

$$a = \left\lceil \sqrt{\frac{w}{h} n} \right\rceil \qquad s_a = \begin{cases} \frac{w}{a} & \text{if } n \leq a \left\lfloor \frac{h}{w} a \right\rfloor \\ \frac{h}{\left\lceil \frac{h}{w} a \right\rceil} & \text{else} \end{cases}$$

Analogously goes the process for determining s_b, whose formula is:

$$b = \left\lceil \sqrt{\frac{h}{w} n} \right\rceil \qquad s_b = \begin{cases} \frac{h}{b} & \text{if } n \leq b \left\lfloor \frac{w}{h} b \right\rfloor \\ \frac{w}{\left\lceil \frac{w}{h} b \right\rceil} & \text{else} \end{cases}$$

Finally, $S = \max \{ s_a, s_b \}$.

Sadly, this algorithm sometimes fails to find the optimal solution. For example, inputs:

- $w = 10, h = 2, n = 8$ expect $S = 1.25$ but this algorithm gives $S = 1$
- $w = 7, h = 2, n = 20$ expect $S = 0.7$ but this algorithm gives $S = 0.(6)$
- and many others

What do these inputs have in common? When does this algorithm have issues?

To investigate this, let's simplify our solution statement. Observe that:

$$\begin{gathered} s_w \in \mathrm{Sol}_a \iff as_w = w \land bs_w \leq h \iff a \geq rb \land s_w = \frac{w}{a_w} \\ s_h \in \mathrm{Sol}_b \iff bs_h = h \land as_h \leq w \iff rb \geq a \land s_h = \frac{h}{b_h} \\ \text{with } n \leq ab \text{ for all of the above} \end{gathered} \tag{9}$$

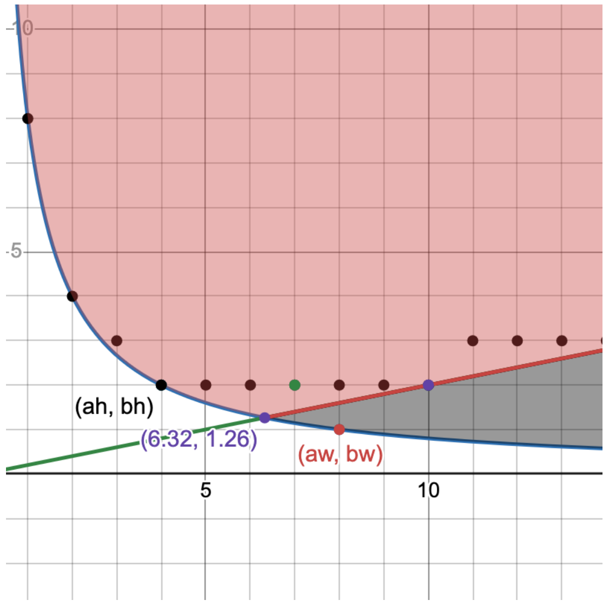
We've simplified width and height to just their ratio, $r = \frac{w}{h}$. Moreover, we don't need the side lengths themselves to compare them. Let $s_w = \frac{w}{a_w}$, $s_h = \frac{h}{b_h}$. Then:

$$s_w > s_h \iff \frac{w}{a_w} > \frac{h}{b_h} \iff \frac{1}{a_w} > \frac{1}{rb_h} \iff a_w < rb_h \tag{10}$$

We can now state the final solution:

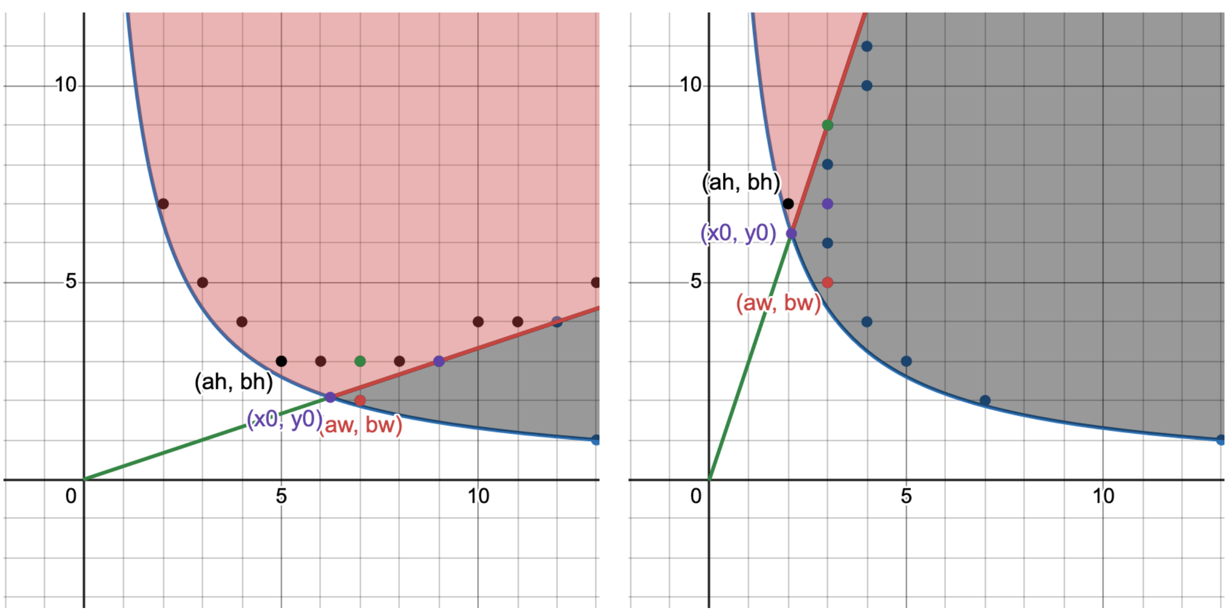
$$\begin{gathered} a_w = \min \left\{ a \in \mathbb{N} : \exists b_w \,.\, a \geq \frac{n}{b_w} \land a \geq rb_w \right\} \\ b_h = \min \left\{ b \in \mathbb{N} : \exists a_h \,.\, b \geq \frac{n}{a_h} \land b \geq \frac{a_h}{r} \right\} \\ s = \begin{cases} s_w = \frac{w}{a_w} & \text{if } a_w < rb_h, \\ s_h = \frac{h}{b_h} & \text{otherwise} \end{cases} \end{gathered} \tag{11}$$

The problem is reduced to just a, b and r. This simplification unlocks a powerful geometric representation in the 2D plane: we're looking for points $(a_w, b_w), (a_h, b_h) \in \mathbb{N}^2$ such that they're on or above the hyperbola $xy = n$ while accounting for their position relative to the line $x = ry$. $x = a$ is used as stand-in for columns in the continuous space, $y = b$ for rows. Take a look (graph A; full visualizer here):



This is the graph for the $n = 8, w = 10, h = 2$ example above, with $r = 10/2 = 5$. The black area under $y = rx$ contains all fit-width solutions, the red area all the fit-height solutions. The union of these two contains the entire solution space, all (a, b) for which $n \leq ab$, i.e. all grid dimensions which fit all squares. Points on the green-red line $x = ry$ are grid dimensions having the ratio exactly r. Black points are a subset of other solution candidates.

The symmetry of the problem with respect to ratio can now be directly visualized. For $n = 13$ and $r = 3$, respectively $r = 1/3$, we have:



Terminology-wise, from here on $(a_w, b_w), (a_h, b_h)$ will be called a *fit-width point*, respectively a *fit-height point*, if the points respect the requirements in (9); if the points are those described in (11), they'll also be called *minimal points* (minimized coordinates for maximal side-length). s_w and s_h will be called *fit-width* and *fit-height solutions*; unless explicitly mentioned, the $(a_w, b_w), (a_h, b_h)$ points implied by s_w and s_h are not necessarily minimal. Sometimes only a_w or b_h will be mentioned but keep in mind that by definition they do have a pair b_w and a_h.

Let's now analyze the solution space.

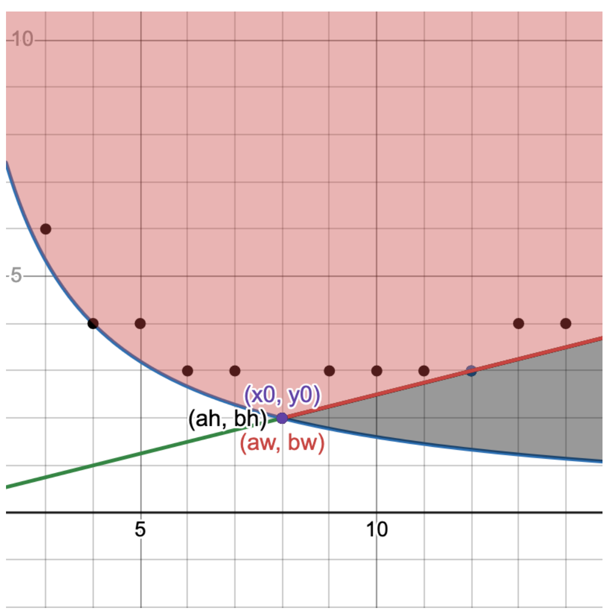
First, notice the purple point at the intersection of the hyperbola and the line – that is, the (x_0, y_0) where $\frac{n}{y_0} = ry_0$ – is special. This would be the "ideal" point, if rows and columns could be fractional somehow: only these dimensions exactly fit all squares and have the exact desired ratio. Notice that:

$$x_0 = \frac{n}{y_0} = ry_0 \iff x_0 = \sqrt{rn} \land y_0 = \sqrt{\frac{n}{r}} \tag{12}$$

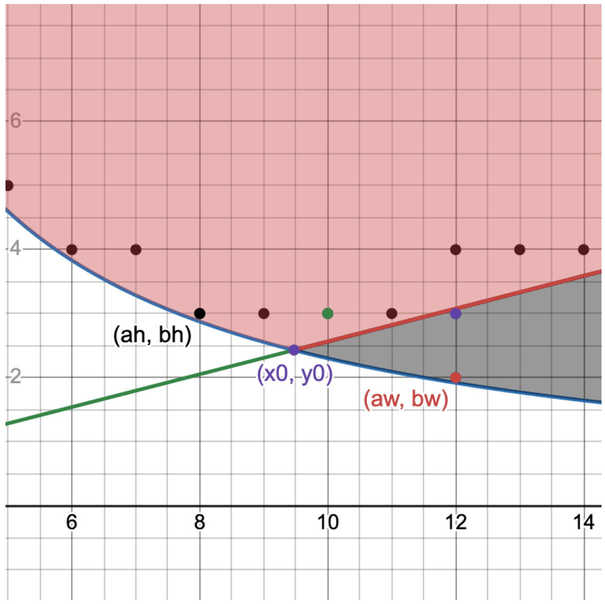
The number of columns a chosen in (2) is precisely $\lceil x_0 \rceil$. This is the visual confirmation of that result: all the fit-width points will find themselves to the left of that point, all the fit-height points to its right. In other words:

$$\forall a_w, b_h \, . \, a_w \geq x_0 \land b_h \geq y_0 \tag{13}$$

If parameters are "nice" such that $x_0, y_0 \in \mathbb{N}$ the minimal fit-width and fit-height points coincide (graph B):



Second, when is a point *valid*, i.e. satisfies solution requirements in (9)? Also, can multiple points give the same solution? Consider the following ($n = 23, r = 3.9$, graph C):



The 4 points next to (a_h, b_h) have the same ordinate and the point above (a_w, b_w) the same abscissa, therefore their corresponding side lengths are still s_h, respectively s_w. This follows directly from the solution requirements in (9):

$$\begin{aligned} &\text{Fit width:} \\ a_w b_w \geq n \wedge a_w \geq r b_w &\iff \frac{n}{a_w} \leq b_w \leq \frac{a_w}{r} \\ &\iff \left\lceil \frac{n}{a_w} \right\rceil \leq b_w \leq \left\lfloor \frac{a_w}{r} \right\rfloor \\ &\text{Fit height:} \\ a_h b_h \geq n \wedge b_h \geq \frac{a_h}{r} &\iff \left\lceil \frac{n}{b_h} \right\rceil \leq a_h \leq \lfloor r b_h \rfloor \end{aligned} \tag{14}$$

We can add the ceiling and floor as a consequence of them being residuated mappings. Notice how the intervals describe the space between the hyperbola and the line for a_h and b_w. Doesn't matter which value from these intervals a_h and b_w have, the solutions of (a_w, b_w), (a_h, b_h) are the same. The ceilings and floors are visible in the graph too: the points are a bit above/under the function graphs too. We can compress the inequality to get the validity condition:

$$\begin{aligned} a_w \text{ valid} &\iff \left\lceil \frac{n}{a_w} \right\rceil \leq \left\lfloor \frac{a_w}{r} \right\rfloor \\ b_h \text{ valid} &\iff \left\lceil \frac{n}{b_h} \right\rceil \leq \lfloor r b_h \rfloor \end{aligned} \tag{15}$$

The number of points with equal solutions is easily derived by subtracting the interval endpoints.

Third, which points give the optimal solution s? How do they compare? Recall (10):

$$s_w > s_h \iff a_w < rb_h$$

From (9) we know that for any (a_h, b_h) fit-height point, $a_h \leq rb_h$, meaning rb_h is an upper bound on a fit-height solution's number of columns. Therefore:

> A fit-width point with less columns than *some* fit-height point gives the better solution between the two.

If $rb_h \notin \mathbb{N}$ then both $a_w \leq \lfloor rb_h \rfloor$ and $a_h \leq \lfloor rb_h \rfloor$, meaning they can have at most as many columns. This is unsurprising: the fit-width solution stretches the columns fully, while the fit-height doesn't.

What about the number of rows? Since $a_w \geq rb_w$ we get that:

$$a_w \geq rb_w \land s_w > s_h \implies rb_w \leq a < rb_h \implies b_w < b_h \tag{16}$$

This means:

> A fit-width point must have less rows than a fit-height point to correspond to a better solution.

It makes sense intuitively: a fit-height solution stretches the rows fully, a fit-width doesn't; if a fit-width solution has as many rows or more, it must be worse.

We can observe these properties on the graphs above:

- in graph A the fit-width solution has less columns: it is the better one
- in graph C the fit-width solution has more columns: it is worse

Graph C shows that having less rows doesn't matter, since one can simply add more rows. Mathematically, for (a_w, b_w) fit-width, (a_h, b_h) fit-height points, if $s_w \leq s_h$:

$$a_w \geq rb_h \geq a_h \land n \leq a_h b_h \implies a_w \geq rb_h \land n \leq a_w b_h \overset{(9)}{\implies} (a_w, b_h) \text{ fit-width}$$

Note that all the results above apply for any r.

Fourth and finally, when $r \geq 1$ if the minimal fit-width solution is strictly better than any fit-height solution, then it corresponds uniquely to a single point (see appendix for proof). Symmetrically for $r < 1$ the same applies for the fit-height solution.

We can now analyze when the algorithm we've developed fails. Translating its solution to points:

$$P_a = \begin{cases} (a_w = \lceil \sqrt{rn} \rceil, b_w = \lfloor \frac{a_w}{r} \rfloor)_0 & \text{if } n \le a_w b_w \\ (a_h = a_w, b_h = \lceil \frac{a_w}{r} \rceil)_1 & \text{else} \end{cases}$$

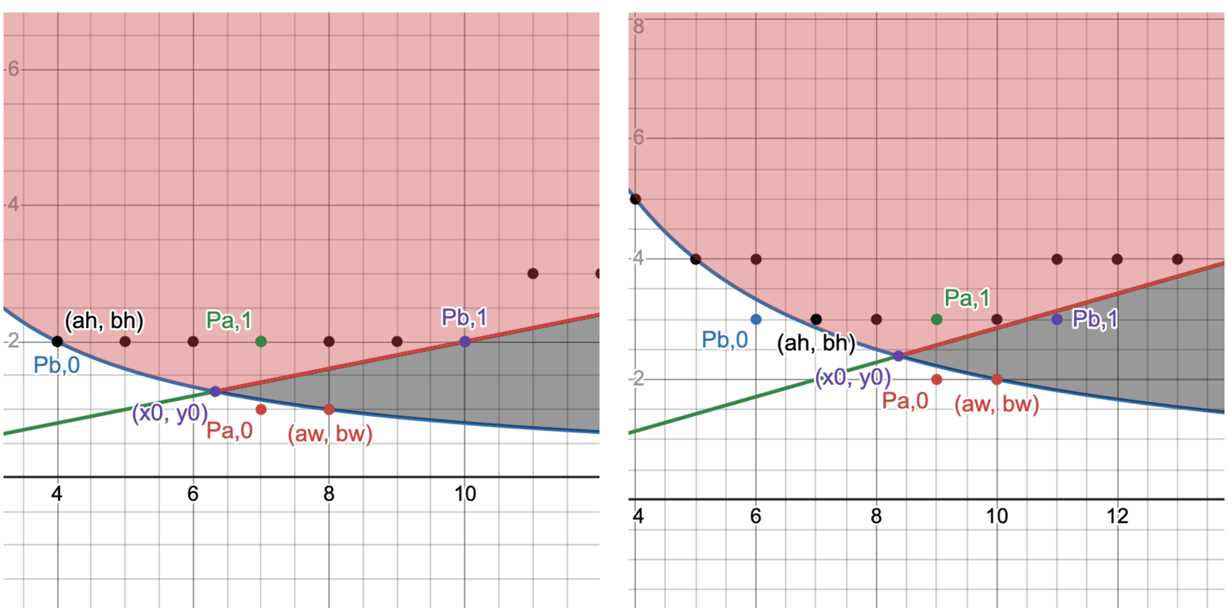
$$P_b = \begin{cases} (a_h = \lfloor r b_h \rfloor, b_h = \lceil \sqrt{\frac{n}{r}} \rceil)_0 & \text{if } n \le a_h b_h \\ (a_w = \lceil r b_w \rceil, b_w = b_h)_1 & \text{else} \end{cases}$$

Recall that if a fit-width point has a solution greater than that of a fit-height point it will have less rows. Above we can see that all of $P_{a,1}, P_{b,0}, P_{b,1}$ have a number of rows greater than y_0: for P_b it is by construction, and for $P_{a,1}$:

$$\left\lceil \frac{a_w}{r} \right\rceil = \left\lceil \frac{\lceil \sqrt{rn} \rceil}{r} \right\rceil \ge \frac{\lceil \sqrt{rn} \rceil}{r} \ge \sqrt{\frac{n}{r}} = y_0$$

Moreover, for $r \ge 1$ there exists a fit-height solution with $b_h = \lceil y_0 \rceil$ with very high probability and fit-width solutions with $a_w = \lceil x_0 \rceil$ are highly unlikely – in fact, almost never likely for $r \ge 2$ (see appendix for proof). Therefore, the optimal solution mainly corresponds to a unique fit-width point (a_w, b_w) with $a_w > \lceil x_0 \rceil$ and $b_w < \lceil y_0 \rceil$ yet the algorithm is optimized to find a fit-height point (a_h, b_h) with $a_h \le \lceil x_0 \rceil$ and $b_h \ge \lceil y_0 \rceil$. By the conclusions in the appendix, the outcome is predominantly ratio-dependent; for the vast majority of ratios the algorithm does not find the optimal solution (e.g. $r \in [1.5, 2)$ paired with most n).

Let's visualize this on the graphs of the two counterexamples given above ($r = 5, n = 8$ and $r = 3.5, n = 20$):



The initial fit-width point candidate $P_{a,0}$ is invalid in both cases (under the hyperbola), so the algorithm falls back to $P_{a,1}$, a fit-height point. In the first example, $P_{b,0}$ is already valid, so the fallback $P_{b,1}$ is not used; in the second, the fallback is used instead but even though it's a fit-width point its solution is equivalent to the fit-height solution, as they have the same number of rows.

Let's now build a new algorithm. This algorithm should find the minimal fit-width and fit height points and pick the solution with the greater side length. Using (15) , the solution we're looking for is:

$$
\begin{aligned}
a_w &= \min \left\{ a \in \mathbb{N} : \left\lceil \frac{n}{a} \right\rceil \leq \left\lfloor \frac{a}{r} \right\rfloor \right\} \\
b_h &= \min \left\{ b \in \mathbb{N} : \left\lceil \frac{n}{b} \right\rceil \leq \lfloor rb \rfloor \right\} \\
s &= \max \left\{ \frac{w}{a_w}, \frac{h}{b_h} \right\}
\end{aligned}
$$

Let's walk through how we translate this to code. First, we are looking for minimum a_w and b_h; from (13) we know that $a_w \geq \lceil \sqrt{rn} \rceil$ and $b \geq \lceil \frac{n}{r} \rceil$. We'll initialize variables `a` and `b` to those values. Then we need to ensure the validity condition; for that, we just increment `a` and `b` in a loop until the condition holds. The loops execute while a_w and b_h are invalid points:

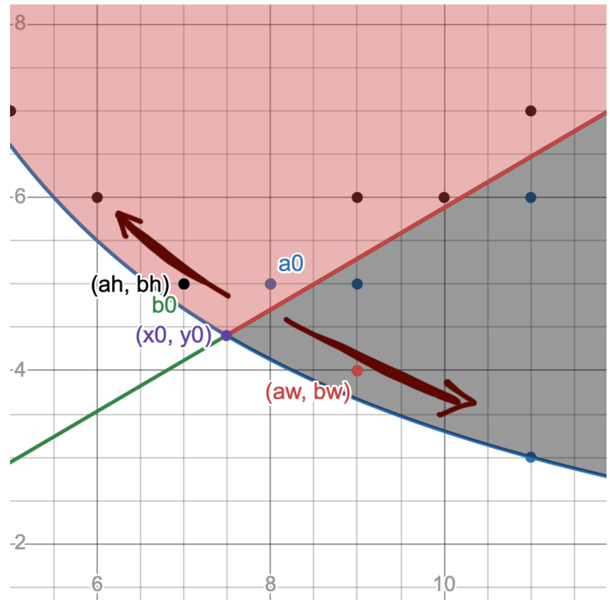
$$
\begin{aligned}
\left\lceil \frac{n}{a} \right\rceil > \left\lfloor \frac{a}{r} \right\rfloor &\iff a < r \left\lceil \frac{n}{a} \right\rceil \\
\left\lceil \frac{n}{b} \right\rceil > \lfloor rb \rfloor &\iff rb < \left\lceil \frac{n}{b} \right\rceil
\end{aligned}
\tag{17}
$$

The implementation follows directly:

```
const r = w / h

let a = Math.ceil(Math.sqrt(r * n))
for (; a < r * Math.ceil(n / a); a++);

let b = Math.ceil(Math.sqrt(n / r))
for (; r * b < Math.ceil(n / b); b++);

return Math.max(w / a, h / b);
```

The algorithm always terminates: as a grows, $\lceil \frac{n}{a} \rceil$ decreases and $\lceil \frac{a}{r} \rceil$ increases, shrinking the interval until the ordering of its endpoints is swapped; likewise for b. The values at the end of each loop are minimal – we check every value one by one in ascending order.

Intuitively, if the loop condition holds there is no valid (a_w, b_w) or (a_h, b_h) fit-width, respectively fit-height point with the iteration's current a_w or b_h value. If we take $b_w = \lceil \frac{n}{a} \rceil$ and $a_h = \lceil \frac{n}{b} \rceil$ (the minimum values for a valid solution) we can imagine this algorithm as "walking away" from (x_0, y_0) alongside the hyperbola until we (inevitably) find ourselves in the fit-width/fit-height solution spaces:

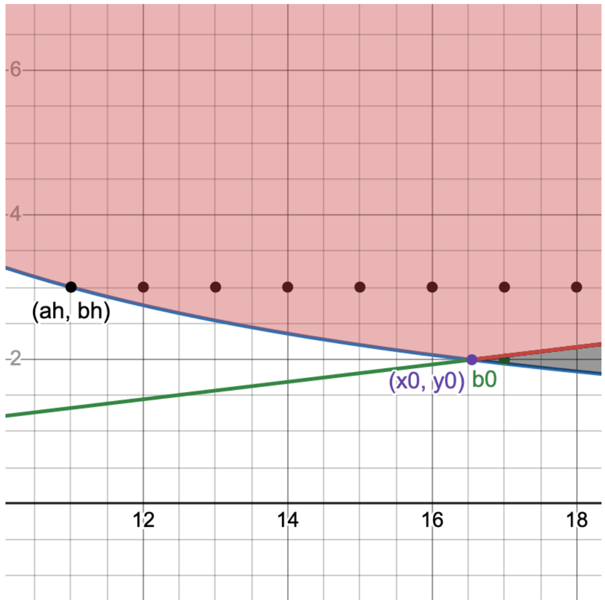


a_0 and b_0 are the starting points, right next to (x_0, y_0). The algorithm "walks down" to (a_w, b_w) and "up" to (a_h, b_h). In fact, in this example, b_0 is initialized directly to (a_h, b_h); a_0 must be incremented once.

We have convinced ourselves of correctness. To analyze its runtime complexity, let's denote $a_0 = \lceil \sqrt{rn} \rceil$ the starting value of a. Since the loop condition is $a < r \lceil \frac{n}{a} \rceil$, a will be incremented at most $\lceil r \lceil \frac{n}{a} \rceil - a_0 \rceil$ times. Since a increases $\lceil \frac{n}{a} \rceil \leq \left\lceil \frac{n}{a_0} \right\rceil$. We can now count iterations:

$$
\begin{aligned}
\left\lceil r \left\lceil \frac{n}{a} \right\rceil - a_0 \right\rceil &< r \left\lceil \frac{n}{a_0} \right\rceil - a_0 + 1 \\
&< r \frac{n}{a_0} + r - a_0 + 1 \\
&= r \frac{n}{\lceil \sqrt{rn} \rceil} + r - \lceil \sqrt{rn} \rceil + 1 \\
&< \frac{rn}{\sqrt{rn}} + r - \sqrt{rn} + 1 \\
&= r + 1
\end{aligned} \tag{18}
$$

This means at most $\lfloor r \rfloor + 1$ iterations and a runtime of $\mathcal{O}(r)$, dependent only on the aspect ratio of the grid. To show that the bound is tight, consider this following example:



Here $n = 33, r = 8.3$. A very rare case when the fit-height starting point b_0 is in the fit-width solution space. Consider the iteration count of the fit-height loop of the algorithm – by the same reasoning it is $\left\lfloor \frac{1}{r} \right\rfloor + 1$. It's visible that in this example the loop will do exactly 1 iteration to reach (a_h, b_h). Since the algorithm is symmetric this proves tightness for the first loop by using the same n and $r' = \frac{1}{r}$.

With a correct algorithm and good understanding of why it works, we conclude here.

Appendix

Uniqueness of fit-width solution when $r \geq 1$

Let (a, b) be a minimal fit-width point. We prove there is no b' such that (a, b') is a fit-width point when $r \geq 1$ and (a, b) corresponds to the best solution, meaning that $s_w > s_h$ for all (a_h, b_h) fit-height points.

Assume towards a contradiction there exists $b' \neq b$ such that (a, b') is a fit-width point. Without loss of generality, we can further assume $b < b'$. From (14) and the minimality of a we can write a in terms of b':

$$n \leq ab' \wedge a \geq rb' \implies a \geq \left\lceil \frac{n}{b'} \right\rceil \wedge a \geq \lceil rb' \rceil \implies a = \max \left\{ \left\lceil \frac{n}{b'} \right\rceil, \lceil rb' \rceil \right\}$$

Suppose that $a = \left\lceil \frac{n}{b'} \right\rceil$. Since we have $n \leq ab$ from (14), $b < b'$ and $r \geq 1$:

$$a - 1 < \frac{n}{b'} \implies b'(a-1) < ab \implies (b' - b)a < b' \implies a < rb'$$

But $a \geq rb'$ so we've reached a contradiction. Therefore, as per the first result:

$$a = \lceil rb' \rceil \implies \left\lceil \frac{n}{b'} \right\rceil < \lceil rb' \rceil \implies \left\lceil \frac{n}{b'} \right\rceil \leq \lfloor rb' \rfloor$$

By (14) this means there is a valid fit-height point (a', b') (real example in graph C). But then, if one takes $s_h = \frac{h}{b'}$:

$$a \geq rb' \implies s_w \leq s_h$$

In conclusion, $b' \neq b$ contradicts $s_w > s_h$, thus b' can't exist and (a, b) is unique.

Position of solution points relative to (x_0, y_0)

We'll approach this subject by handling fit-height points where $r \geq 1$. Due to the symmetry of the problem relative to ratio, we can make more general conclusions afterwards.

$b = \lceil y_0 \rceil = \left\lceil \sqrt{\frac{n}{r}} \right\rceil$ is a valid fit-height point by (15) if and only if:

$$\left\lceil \frac{n}{b} \right\rceil \leq \lfloor rb \rfloor$$

A sufficient condition is:

$$\begin{aligned} rb - \frac{n}{b} \geq 1 &\implies rb^2 - n - b \geq 0 \\ &\implies b \geq \frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} > \sqrt{\frac{n}{r}} \end{aligned}$$

If b is greater than that the fit-height point (a, b) is valid. The condition excludes cases when $\exists k \in \mathbb{N} \,.\, k \in [\frac{n}{b}, rb]$ (e.g. when $y_0 \in \mathbb{N}$ then the point valid for every n) but it is easy to *quantify*. We can use it to answer: for fixed $r \geq 1$ what's the probability for b to be a valid fit-height solution?

To do this, observe that by the properties of ceiling, $\sqrt{\frac{n}{r}} \leq b < \sqrt{\frac{n}{r}} + 1$. We can bound the expression above likewise, using $r \geq 1$:

$$\sqrt{\frac{n}{r}} < \frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} < \frac{1}{2} + \sqrt{\left(\frac{1}{2} + \sqrt{\frac{n}{r}}\right)^2} = \sqrt{\frac{n}{r}} + 1$$

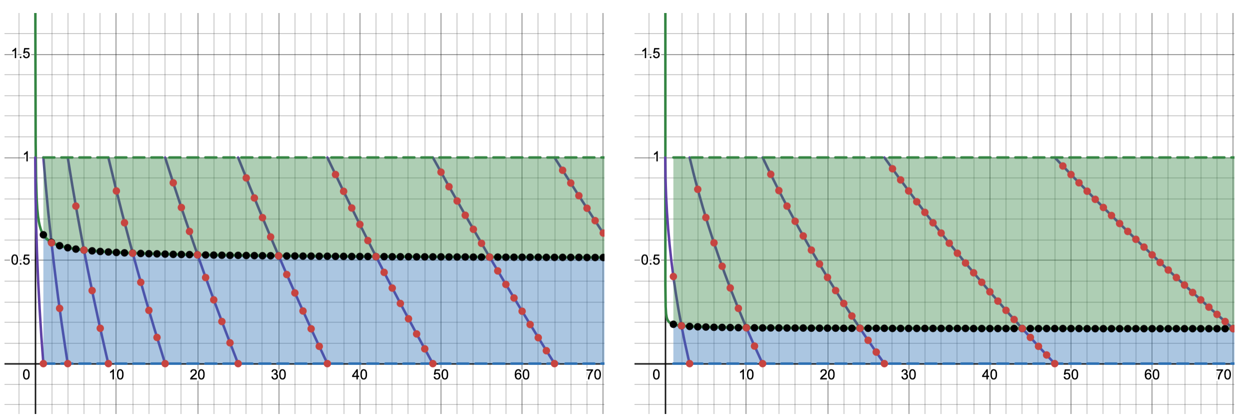
It follows that:

$$\sqrt{\frac{n}{r}} < \frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} \leq b = \left\lceil \sqrt{\frac{n}{r}} \right\rceil < \sqrt{\frac{n}{r}} + 1$$

If we subtract $\sqrt{\frac{n}{r}}$, the difference will be always between 0 and 1:

$$0 < \underbrace{\frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} - \sqrt{\frac{n}{r}}}_{B_r(n)} \leq \underbrace{\left\lceil \sqrt{\frac{n}{r}} \right\rceil - \sqrt{\frac{n}{r}}}_{C_r(n)} < 1$$

Let's visualize this:



First graph depicts $r = 1$, the second $r = 3$. The red points represent $C_r(n)$, the black points $B_r(n)$. Whenever a red point is in the green area $C_r(n) \geq B_r(n)$, meaning $\left\lceil \sqrt{\frac{n}{r}} \right\rceil$ is a valid fit-height point for n.

Since this criterion doesn't cover all possibilities, the ratio between the number of valid points in the green area and the total number of points provides a lower bound for the probability that b is a valid point:

$$P_r \geq \lim_{n \to \infty} \frac{|S_n|}{n} \qquad S_n = \{\, k \in [n] : C_r(k) \geq B_r(k) \,\}$$

We can approximate by counting points up to a certain n. For example $P_{1,n=10} \geq \frac{4}{10}$ and $P_{3,n=26} \geq \frac{21}{26}$.

But we can't count to infinity. Enter asymptotic density: a measure of how big a subset of $\mathbb{N}$ is relative to $\mathbb{N}$! For our solution set S its density is:

$$d(S) = \lim_{n\to\infty} \frac{|S_n|}{n} \le P_r$$

Exactly what we need. To compute $d(S)$ we use that $C_r(n)$ is equidistributed modulo 1. Equidistribution modulo 1 implies that, for constant $t \in [0, 1)$:

$$\lim_{n\to\infty} \frac{|\{\, C_r(k) : k \in [n] \,\} \cap [t, 1]|}{n} = 1 - t$$

This tells us that a percentage of $1 - t$ points out of n are in $[t, 1]$. One can roughly notice this on the graph: the points $(n, C_r(n))$ are distributed uniformly between the green and blue areas, proportionally with their sizes.

The limit's numerator is equivalent to $|\{\, k \in [n] : C_r(k) \ge t \,\}|$. For $t = B_r(k)$ this would've been precisely S_n, with the limit then giving us $d(S)$ directly, but t must be constant. To find an appropriate value, let's bound $B_r(n)$:

$$\begin{aligned}
\frac{1}{2r} \le B_r(n) &= \frac{1}{2r} + \sqrt{\frac{1}{4r^2} + \frac{n}{r}} - \sqrt{\frac{n}{r}} \\
&= \frac{1}{2r} + \sqrt{\frac{n}{r}}\left(\sqrt{1 + \frac{1}{4rn}} - 1\right) \\
&\le \frac{1}{2r} + \sqrt{\frac{n}{r}} \cdot \frac{1}{8rn} \\
&= \frac{1}{2r} + \frac{1}{8r\sqrt{rn}}
\end{aligned}$$

The approximation $\sqrt{1+x} \le 1 + \frac{x}{2}$ was used. Observe that the second term vanishes as $n \to \infty$. There may be an initial finite interval on which it excludes numbers, but for the remaining infinite tail it changes nothing. Since the density of a finite subset of integers is 0, we can ignore that segment when computing $d(S)$. The effect is also visible on the graph early on.

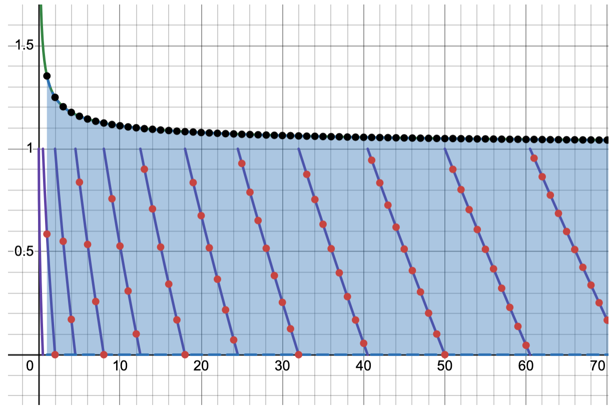
This means we can simply take $t = \lim_{n\to\infty} B_r(n) = \frac{1}{2r}$. Our final result is:

$$P_r \ge d(S) = \lim_{n\to\infty} \frac{\left|\{\, C_r(k) : k \in [n] \,\} \cap [\frac{1}{2r}, 1]\right|}{n} = 1 - \frac{1}{2r}$$

As a final note, attempting to exclude points where $C_r(n) = 0$, i.e. $\sqrt{\frac{n}{r}} \in \mathbb{N}$, would've not influenced the result. $\frac{n}{r} \in A$, where $A = \left\{\, p^2 \;\middle|\; p \in \mathbb{N} \,\right\}$ the set of all squares, and since $|A \cap [n]| \le \mathcal{O}(\sqrt{n})$, $d(A) = 0$. Our set of points is a subset of that, so it must also be density-zero. By other means one could strengthen the bound to $1 - \frac{1}{4r}$ for $r \in \mathbb{R} \setminus \mathbb{Q}$, and to $p \in [1 - \frac{1}{2r}, 1]$ otherwise (with $p = 1$ for $r \in \mathbb{N}$).

Thus, for the examples above $P_1 \geq 0.5$ and $P_3 \geq 0.833$. Desmos for the graph here.

The first conclusion is that fit-height solutions will for the most part have $b_h = \left\lceil \sqrt{\frac{n}{r}} \right\rceil$. We can derive even more interesting information by looking at the graph for $r < 1$:



Note that $r < 1$ means we've essentially flipped dimensions: the fit-height points correspond to the fit-width points of ratio $\frac{1}{r}$. For $r = \frac{1}{2}$ as depicted above, $P_{\frac{1}{2}} \geq 0$. How do we interpret this?

> For $r \geq 2$, no a = $\lceil \sqrt{rn} \rceil$ satisfies the sufficient condition above to fulfill the fit-width solution requirements.

Any valid a_w will be greater than $\lceil x_0 \rceil$ with very high certainty, especially when considering the growth rate of $\frac{1}{2r}$ as $r \to 0$. This also provides an intuitive explanation for the second algorithm's runtime: as r grows the fit-width solutions are farther and farther away from (x_0, y_0), so the first loop will have to iterate more; at the same time, the probability that the initial fit-height solution candidate is valid rapidly increases, which prevents the second loop from iterating.

The final conclusion is: for $r \geq 1$ it is very likely that $b_h = \left\lceil \sqrt{\frac{n}{r}} \right\rceil$ is a valid fit-height point and almost never likely that $a_w = \lceil \sqrt{rn} \rceil$ is a valid fit-width point. Then mostly $b_w < b_h$, since with more columns the grid requires less rows to fit all squares. Therefore by (17) for $r \geq 1$ *the fit-width solution is in the majority of cases better than any fit-height solution*. Likewise for $r < 1$ but with fit-width and fit-height swapped.

Using binary search in the second algorithm

According to (18) the algorithm always finds a solution after $\lfloor r \rfloor + 1$ iterations, thus $a_0 \leq a_w < a_0 + r + 1$, where $a_0 = \lceil \sqrt{rn} \rceil$ is the starting value for a_w.

We can binary search over that interval using the same loop condition from (17). If the midpoint is not a valid a_w, we search in the upper half, since we're not yet in the solution space, otherwise we search in the lower half, since we want the minimal solution. Here is the algorithm for a_w:

```
const r = w / h

let left = Math.ceil(Math.sqrt(r * n))
let right = Math.ceil(left + r + 1)

while (left < right) {
    const mid = Math.floor(left + (right - left) / 2)
    if (mid < r * Math.ceil(n / mid)) {
        left = mid + 1
    } else {
        right = mid
    }
}

return w / left
```

Upper bound ceiled so interval is open. `left` stores a_w. This reduces the asymptotic runtime to $\mathcal{O}(\log r)$. Useful for very large ratios but very large ratios seem utterly useless.

$\lfloor\cdot\rfloor$ and $\lceil\cdot\rceil$ are residuated mappings

This means that, for $\forall x \in \mathbb{R}, n \in \mathbb{N}$:

1. $n \leq x \iff n \leq \lfloor x \rfloor$
2. $x \leq n \iff \lceil x \rceil \leq n$
3. $n < x \iff n < \lceil x \rceil$
4. $x < n \iff \lfloor x \rfloor < n$

To prove the first, if $x \in \mathbb{N}$ then clearly $n \leq \lfloor x \rfloor = x$, else if $x \in \mathbb{R} \setminus \mathbb{N}$:

$$
\begin{aligned}
n \leq x \iff n < x &\iff n < \lfloor x \rfloor + \{x\} \\
&\iff n \leq \lfloor x \rfloor \vee (\lfloor x \rfloor < n < \lfloor x \rfloor + \{x\}) \\
&\overset{\{x\}<1}{\iff} n \leq \lfloor x \rfloor \vee (\lfloor x \rfloor < n < \lfloor x \rfloor + 1) \\
&\overset{n\in\mathbb{N}}{\iff} n \leq \lfloor x \rfloor
\end{aligned}
$$

The others are proven in a similar fashion. Read more about this on Wikipedia's "Equivalences" for floor and ceiling. These are standard properties but I insisted on writing them here as a "note to self" since I've confused myself and applied them wrong way too many times. I probably know that Wikipedia page now by heart.